Contents

A Sector-Tilt Signal That Survived Its Own Screening

Macro liquidity and the Tech-vs-Consumer relative trade, 2008–2026

Tim Wang · timothy19won@gmail.com · github.com/Timothy-yqw19/qdii-sector-rotation

---

Executive summary

I built a three-dimension scoring framework for the US tech-vs-consumer sector pair, then ran it through a screening process designed to kill it. **Two of the three dimensions failed and were removed. The third survived.**

What remains is a monthly macro-liquidity tilt with a 20-day Spearman IC of **0.189**, a Newey-West *t* of **3.14** that stays above 3 even with a full year of HAC lags, and an information ratio of **0.43** against a neutral 50/50 benchmark at 1.7% tracking error.

It also has a defect I could not engineer away: **strip out the best 12 months of 236 and the cumulative excess return collapses from +15.2% to +1.9%.** The signal is not an all-weather timing tool. It is a *stress-period* signal — it earns +1.41% annualised during risk-off regimes versus +0.41% in calm markets.

That reframing is the actual conclusion of this project, and it only emerged because the diagnostics were built to look for it.

---

1. Why this pair, and why relative

Universe. Long leg: hard tech / AI (QQQ, with SMH and XLK as signal proxies). Short leg: consumer discretionary (XLY, with XRT). Benchmark SPY.

Relative, not absolute. Absolute timing requires calling both direction and magnitude; relative strength only requires ranking two assets. Common market beta cancels, which raises signal-to-noise and matches the actual decision a sector allocator faces. A consequence worth stating: any factor that moves both sectors equally nets out of the spread by construction. That is correct behaviour, not a missing feature.

Three dimensions, chosen before looking at results:

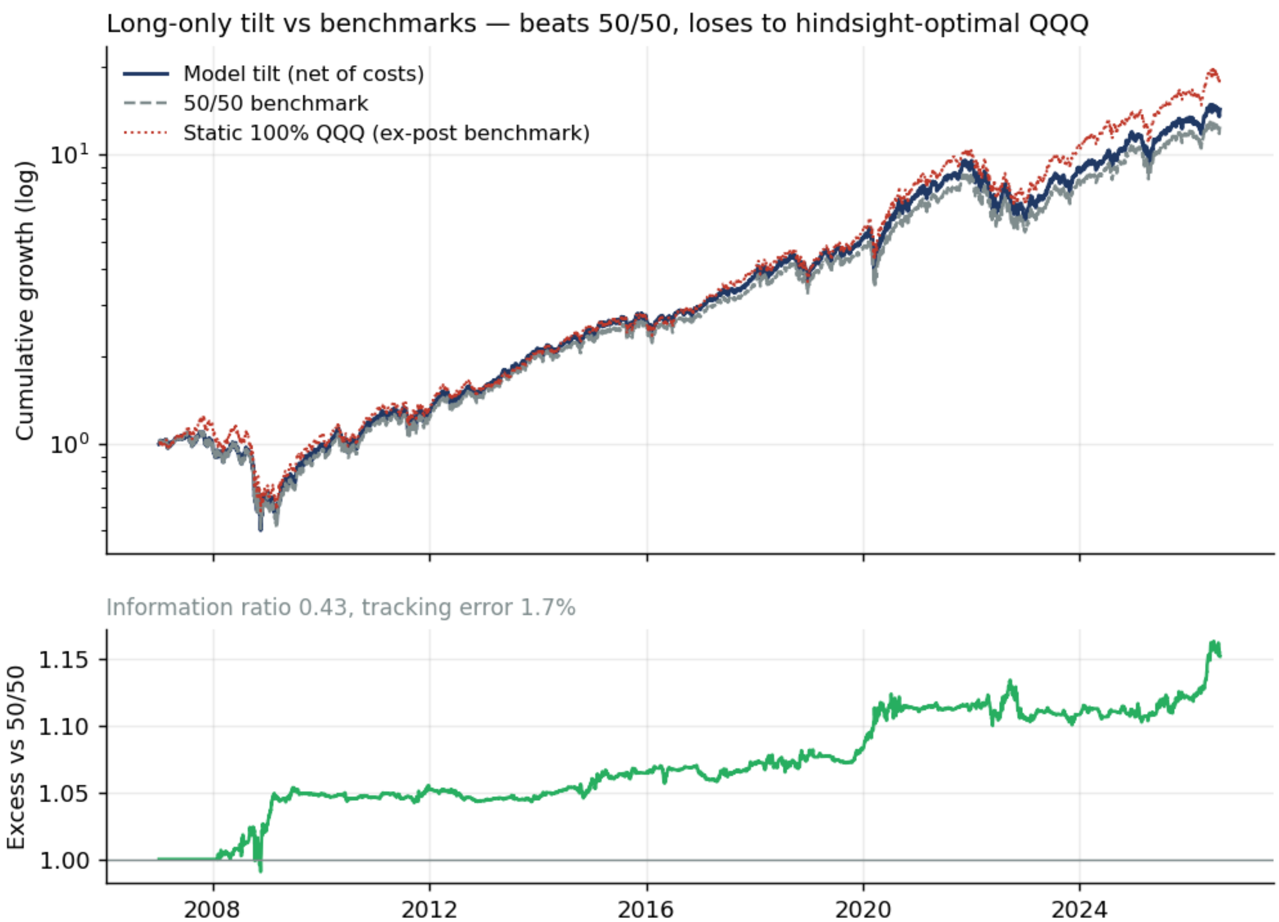

Figure 1: Equity curve

Dimension	Speed	Signals
Momentum & positioning	Fast (daily)	Relative momentum 20d/60d, vol-adjusted momentum, money-flow proxy
Macro liquidity & risk appetite	Mixed	Real 10y rate, broad dollar, credit spread, Fed balance sheet, VIX
Relative valuation	Slow	Relative price percentile, distance from 200d MA percentile

Each signal carries an explicit direction prior with a written economic rationale, and the *magnitudes* differ across the two sectors — that asymmetry is where the relative signal actually comes from. Example: a stronger dollar hurts tech more than domestic-facing consumer names, because Nasdaq-100 foreign revenue runs roughly 50–60% versus roughly 15% for XLY.

---

2. The screening: two dimensions rejected

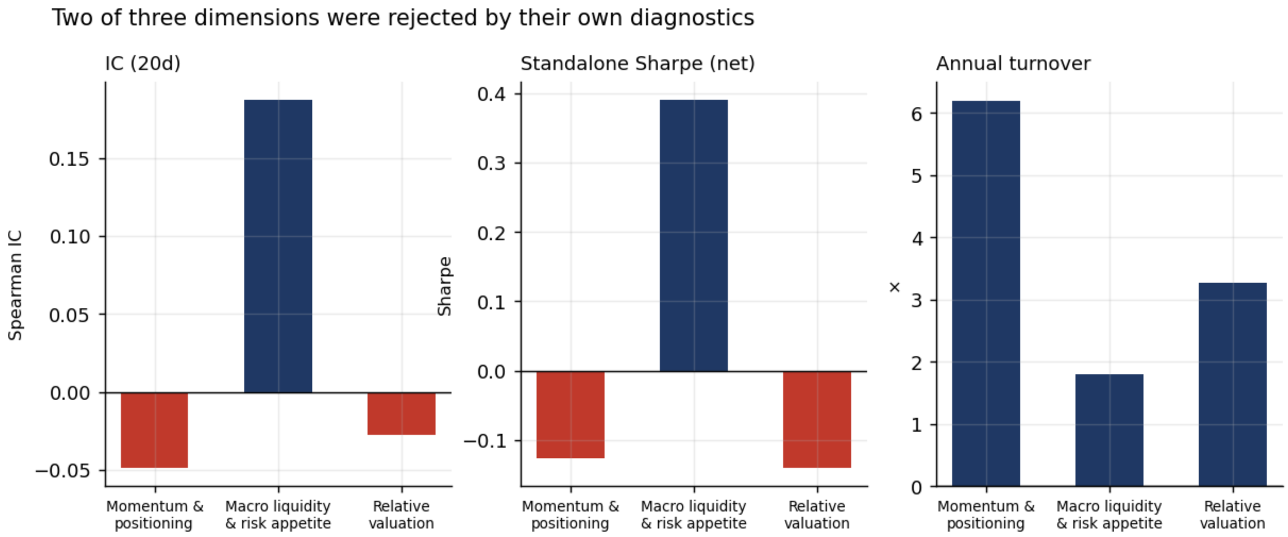

Figure 2: Dimension screening

Dimension	IC 5d	IC 20d	IC 60d	Standalone Sharpe	Turnover	Max DD	Verdict
Momentum & positioning	-0.024	**-0.049**	-0.034	**-0.126**	6.20	-34.2%	**Rejected**
Macro liquidity	+0.085	**+0.187**	**+0.209**	**+0.390**	1.81	-6.8%	**Kept**
Relative valuation	-0.003	-0.028	-0.054	-0.140	3.26	-34.1%	**Rejected**

Momentum was not merely uninformative — it was actively destructive: 3.4× the turnover of the macro dimension, 5× the drawdown, negative return. Under daily rebalancing it was worse still (Sharpe -0.45, turnover 25.0×). Equal-weighting all three diluted the macro dimension's +0.187 down to **-0.045** and dragged the Newey-West t from +3.14 to **-0.59**.

2.1 A citation I got wrong

I originally justified the momentum dimension with Moskowitz & Grinblatt (1999), *Do Industries Explain Momentum?* **That citation was misapplied.** Their result is cross-sectional momentum across roughly 20 industry portfolios — a ranking effect across a broad cross-section. This project runs *pairwise* relative momentum between two highly correlated sectors, where short-horizon mean reversion is the more common empirical regularity. The negative IC is internally consistent; it is not a data problem.

The broader lesson is about transferability: an effect documented in one cross-sectional setting does not automatically survive translation to a two-asset pair. Checking that translation should have happened before the signal was built, not after.

2.2 The surviving dimension is not one variable in disguise

Signal	IC 20d	2008–14	2014–20	2020–26
Fed balance sheet, 60d change	0.120	0.157	0.286	0.029
Credit spread (Baa–10y), 20d change	0.118	0.084	0.187	**0.107**
Broad dollar, 60d change	0.105	-0.046	0.252	0.076
Real 10y rate, 60d change	0.053	0.136	0.051	-0.035
VIX, 20d change	0.038	0.055	0.051	0.020
Composite (equal-weighted)	**0.187**	**0.162**	**0.336**	**0.104**

Two observations.

The composite beats every individual signal (0.187 versus a best single of 0.120). Diversification is genuinely reducing idiosyncratic noise rather than stacking correlated bets — which is the justification for keeping the whole dimension instead of trading the Fed balance sheet alone.

My second prior was also wrong. I expected the real rate to dominate, on standard equity-duration logic: tech cash flows sit further out, so they should be more discount-rate sensitive. It is the weakest of the five and flips sign in the most recent sub-sample. The work is being done by the Fed balance sheet and the credit spread.

---

3. Does it survive stress-testing?

3.1 Sub-sample stability

Sub-period	IC 20d	Hit rate	Net Sharpe	Realised tech-vs-consumer drift
2008-01 – 2014-04	0.156	56.2%	0.358	**-2.0%/yr**
2014-04 – 2020-06	0.334	62.6%	0.755	+5.0%/yr
2020-06 – 2026-08	0.114	55.0%	0.264	+7.2%/yr

The obvious alternative explanation for any tech-vs-consumer signal over this window is that it is one long bet on falling real rates and a secular tech bull market. The sub-sample split addresses this directly: **the IC is positive in the one sub-period where tech actually underperformed** (2008–14, drift -2.0%/yr). That is not what a disguised buy-and-hold would produce.

The caveat is equally clear: the middle period is much stronger than the other two, and the most recent one is the weakest.

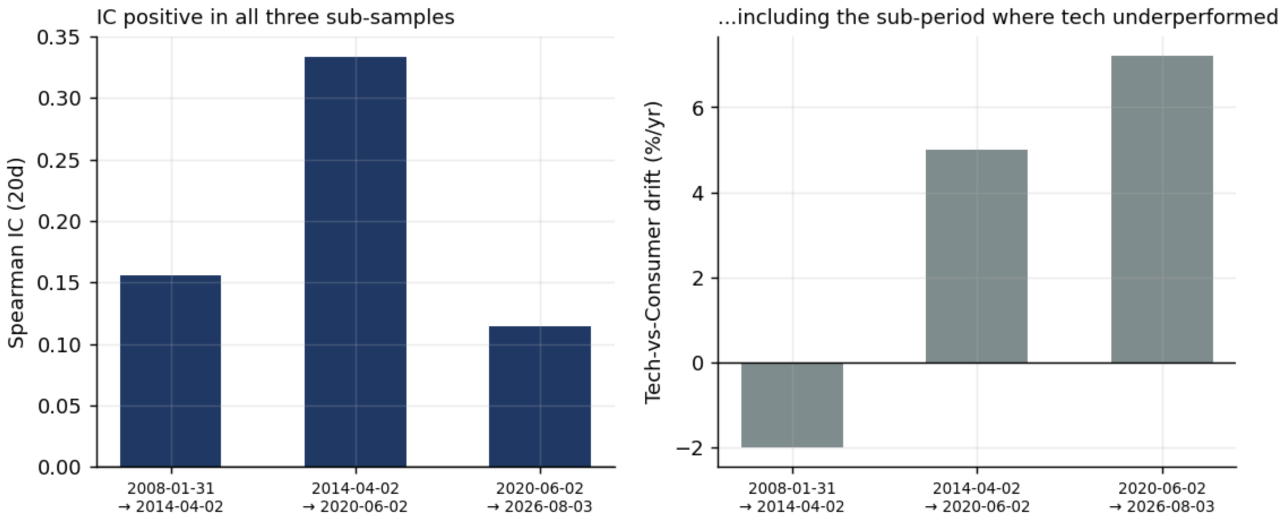

Figure 3: Sub-periods

3.2 Significance after correcting for autocorrelation

Slow-moving signals with overlapping forward windows produce heavily autocorrelated residuals; naive *t*-statistics overstate significance systematically. Rather than acknowledging this in prose, the report shows the full decay curve.

```
20-day horizon: t_naive = 11.83 → L=22: 3.15 → L=63: 2.90 → L=126: 3.04 → L=252: 3.14
 5-day horizon: t_naive =  5.45 → L=7:  2.63 → L=63: 2.59 → L=126: 2.74 → L=252: 2.68
```

Extending the HAC lag window to a **full trading year** leaves t at 3.14. The naive 11.83 is the number to discard. Signal persistence: $\rho_1 = 0.997$, autocorrelation half-life ≈ 215 trading days (10.2 months), and 86 sign changes over 18.4 years — roughly **4.7 independent bets per year**.

> A note on what *not* to report: the code also computes an AR(1) effective sample size, which returns N_eff ≈ 7. That number is not credible — the formula $N(1-\rho)/(1+\rho)$ collapses mechanically as $\rho \to 1$. It is flagged as a lower bound in the output rather than quoted as a result.

3.3 Where the signal actually works

Quintile	Mean 20d forward relative return	Share positive
1 (lowest)	-0.15%	45.2%
2	-0.21%	47.3%
3	-0.10%	50.1%
4	+0.43%	59.9%
5 (highest)	**+1.52%**	**70.6%**

The payoff is **asymmetric**. Nearly all of it sits in the top quintile; the bottom three are mildly negative and barely distinguishable from each other. This is an overweight-confirmation signal. There is no evidence supporting the short side, and it should not be traded that way.

3.4 Weighting: three schemes, no hand-picked numbers

Signals are averaged within each dimension (so that four correlated momentum measures do not vote four times), then combined across dimensions. Rather than asserting a weight vector, three

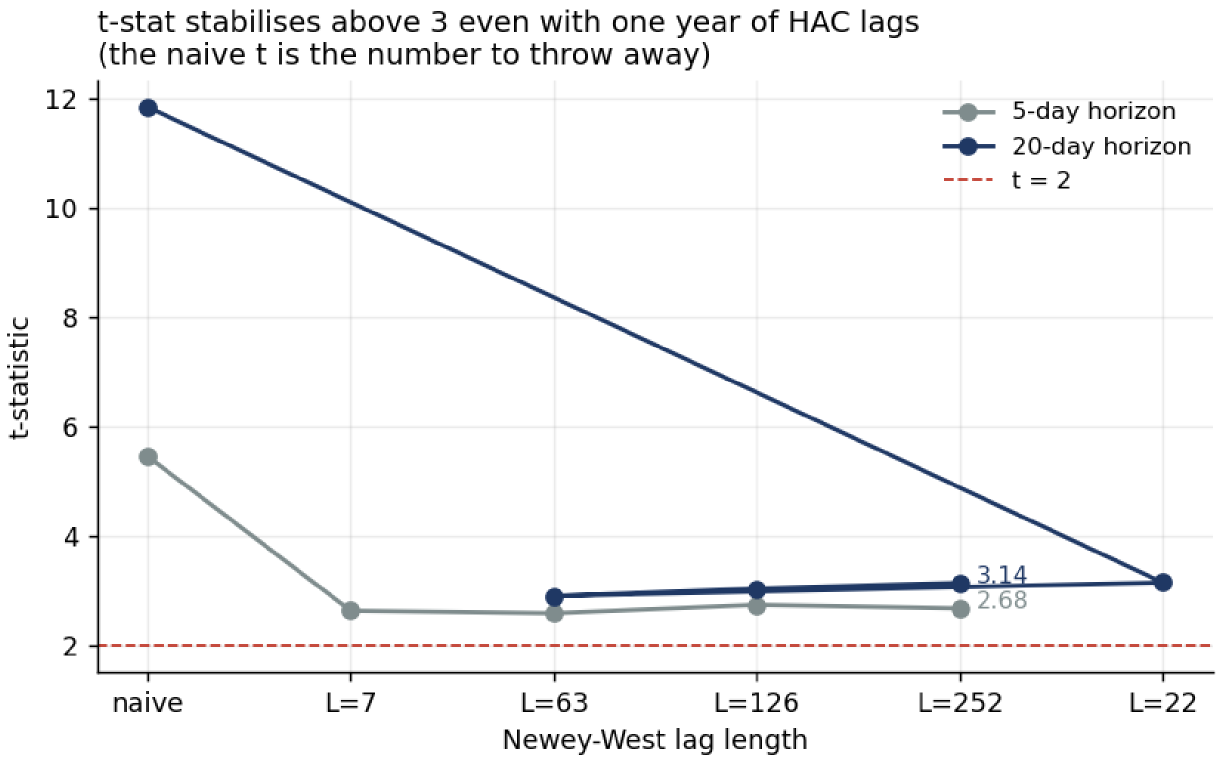

Figure 4: Newey-West decay

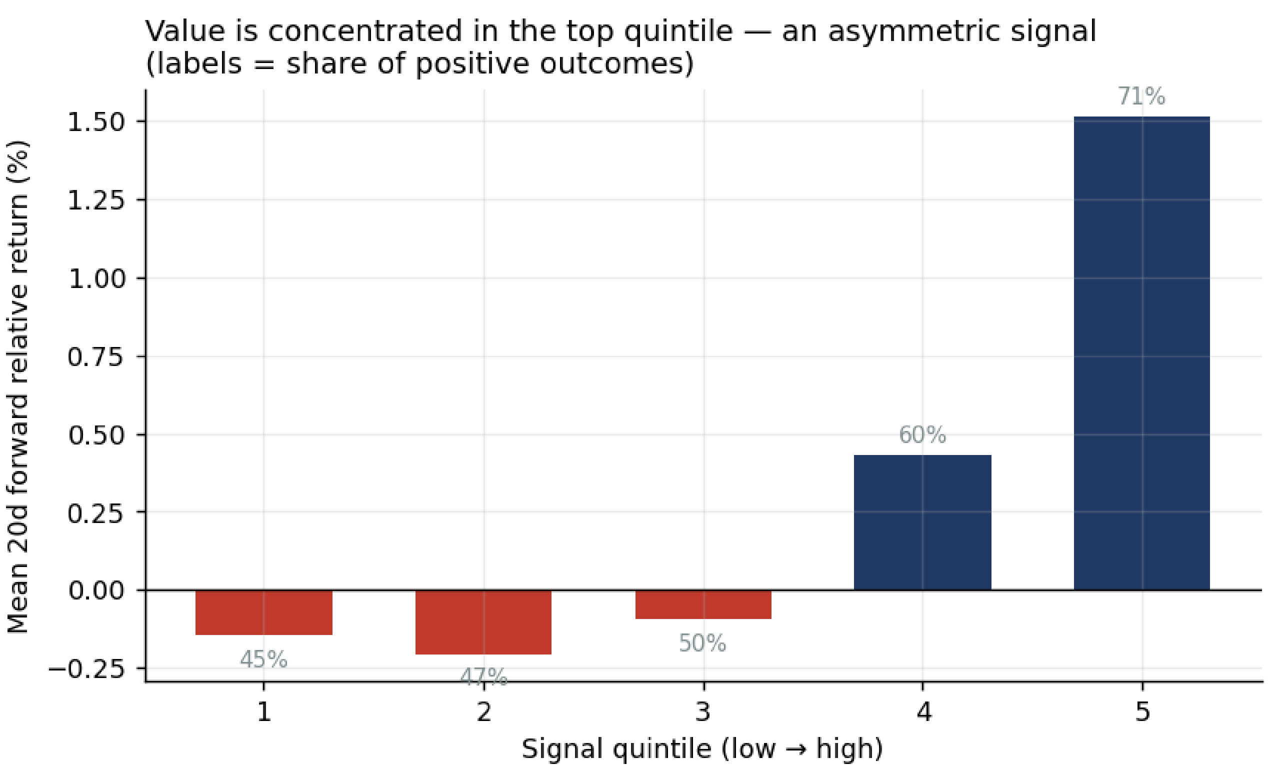

Figure 5: Quintiles

schemes are run under identical conditions:

Scheme	IC 20d	Net Sharpe	IR vs benchmark	Turnover
Equal (adopted)	0.1894	**0.425**	**0.432**	1.61
Inverse volatility	0.1912	0.419	0.426	1.50
Rolling IC-weighted (shrunk)	0.1870	0.358	0.364	1.43

They are effectively indistinguishable, and equal weighting wins. IC weighting — standard practice on the sell side — adds nothing here. A 300-draw Dirichlet sensitivity analysis is also available.

The implementation detail that matters most: **IC weights are estimated only from fully realised forward returns.** The rolling IC series is shifted forward by `horizon + execution_lag` − 1 days, so the most recent observation feeding the weight at time *t* has an outcome that was already known at *t*. One day less would be look-ahead, and two dedicated tests guard this.

---

4. Regime switching: proposed, tested, rejected

Fixed direction priors assume the macro-to-sector transmission is invariant. That is questionable — easing in a recession is a *response to* a crisis, not a tailwind.

Method. Risk-on/risk-off classified from rolling percentiles of the credit spread and VIX, with hysteresis thresholds (enter 0.70, exit 0.50). Result: risk-on 68.3% / risk-off 31.7%, ~20 episodes each, mean duration 160 / 78 trading days. Crucially, the risk-off multipliers were **declared in advance from economic reasoning and never fitted**, because roughly 20 risk-off episodes cannot support estimating five coefficients.

Variant	IC 20d	Net Sharpe	IR	Max DD	Turnover
Baseline (kept)	**0.189**	**0.425**	**0.432**	-6.6%	1.61
Regime-switched priors	0.181	0.300	0.309	-8.3%	1.79
Regime-scaled exposure (0.5×)	0.189	0.408	0.413	**-3.8%**	1.45

Verdict: not adopted. The acceptance criterion was fixed before running — both IC and net-of-cost Sharpe had to improve materially. Both got worse.

Why it failed is more interesting than that it failed. Three of five economic priors were directionally right; two were backwards:

Signal	My prior	IC risk-on	IC risk-off	Verdict
Credit spread	amplify 1.5×	0.100	**0.146**	correct
VIX	amplify 1.5×	0.018	**0.072**	correct
Dollar	amplify 1.3×	0.102	**0.133**	correct
Fed balance sheet	damp 0.5×	0.083	**0.169**	**backwards**
Real rate	damp 0.3×	0.066	0.039	over-damped

Risk signals do bite harder under stress, as expected. But the intuition that "balance-sheet expansion during stress is reactive and therefore contaminated" is **empirically inverted** — its IC in risk-off is double its IC in risk-on. In hindsight this is obvious: the 2009 and 2020 QE announcements *were* the turning points for growth equities. Damping that signal discarded the most valuable part of it.

I did not recalibrate the multipliers to the observed regime-conditional ICs. That would be precisely the overfitting the pre-declaration was designed to prevent, on ~20 episodes. The output of this experiment is a falsified hypothesis and a corrected economic intuition — not a new parameter set.

One finding worth keeping: regime-scaled exposure **halves the maximum drawdown (-6.6% → -3.8%) at a cost of 0.017 in Sharpe.** For a real portfolio that is often a good trade. It is documented as a risk-management option rather than folded into the default model.

---

5. What is wrong with this model

Stated up front rather than waiting to be asked.

1. It does not beat buying and holding QQQ (excess Sharpe -0.25). The defence is that `static_long_leg` is an *ex-post* benchmark — nobody knew in 2008 that tech would dominate — and against the ex-ante defensible 50/50 the IR is 0.43 at 1.7% tracking error. But the defence has a limit: **if an investor's real alternative is simply owning QQQ, this model adds nothing for them.**

2. Returns are extremely concentrated.

Excluding the best…	Remaining cumulative excess
nothing (236 months)	+15.2%
3 months	+10.2%
6 months	+6.6%
12 months (5% of sample)	**+1.9%**
24 months (10%)	**-4.5%**

Monthly hit rate is 53.8%. Split by regime: **+1.41% annualised excess in risk-off (Sharpe 0.58) versus +0.41% in risk-on (Sharpe 0.35)**. This is a stress-period signal. That is coherent with the regime diagnostics in §4 — but it means the effective sample shrinks to a handful of crises, and future performance depends on the next stress episode resembling the last few.

3. Only the overweight side is supported (§3.3).

4. About 4.7 independent bets per year. Enough to support $t \approx 3$, nowhere near enough for a strong claim, and it implies wide dispersion in any few-year realisation.

5. Decay. IC by sub-period runs 0.156 → 0.334 → 0.114. Whether this is alpha decay or the 2020s AI narrative overwhelming macro is unresolved; either way, current-period application deserves a haircut.

Other known limitations. The valuation dimension uses a relative-price-extension proxy rather than true forward P/E (free sources do not carry long forward-P/E history; a user-supplied `data/pe_history.csv` switches it over). The flow signal is a price-volume construct, not actual ETF creations/redemptions. The long-short variant excludes borrow and market-impact costs.

---

6. Method: making the result falsifiable

Anti-look-ahead — 13 automated tests. The strongest is **truncation invariance**: truncate the dataset at any historical date, re-run the whole signal chain, and every value before the cut must match the full-sample run point for point. Any leakage — full-sample standardisation, forward-filled future data, a mis-signed shift — fails it. Additional tests cover rolling z-score and percentile causality, macro publication lags, forward-return alignment, regime-classification causality, a shuffled-signal null (IC ≈ 0), and a perfect-foresight control that *must* be highly profitable or the backtest engine is wired backwards.

Timing convention. Signal computed at close T → executed at close T+1 → returns accrue from T+2 (`execution_lag=2`). Macro series are lagged by their real publication delay (+1 day for daily market series, +4 for the broad dollar index, +9 for the Fed balance sheet), and the order — lag, then align to trading days, then difference — is not interchangeable.

Data. Prices from Tiingo (dividend- and split-adjusted, 2007–present), with Stooq and yfinance as fallbacks; no cross-source mixing within a run, since inconsistent adjustment between the two legs would corrupt the relative return. Macro from FRED.

One data incident worth recording. The credit-spread series was originally `BAMLH0A0HYM2` (ICE BofA high-yield OAS), which returned only 787 rows starting 2023-08. I diagnosed silent endpoint truncation and wrote multi-path retry logic. The actual cause was in the FRED page notes: *"Starting in April 2026, this series will only include 3 years of observations"* — an ICE Data licensing change. No amount of retrying could have fixed it. Substituting `BAA10Y` (Moody's Baa minus 10-year Treasury), which measures the same credit risk premium with 40 years of unrestricted history, raised the most recent sub-period IC from 0.071 to 0.104. The lesson now sits in the code comments: check the data source's own documentation before writing code against an assumed failure mode.

---

7. How I would actually use it

Not as a timing engine. As a **monthly, asymmetric, stress-period overweight confirmation signal**:

- Act on the top quintile; ignore the bottom (§3.3).
- Expect it to contribute during risk-off episodes and contribute little otherwise (§5.2).
- Size it as one input among several, not as a standalone strategy — an IR of 0.43 at 1.7% tracking error is a modest tilt, not a return engine.
- Re-examine it if the risk-off IC advantage disappears, which is the specific condition under which the thesis breaks.

The strongest claim I am willing to make is narrow: *over 2008–2026, in this sector pair, a diversified macro-liquidity composite carried information about one-month relative returns that survived HAC correction, sub-sample splits, and a weighting-scheme sensitivity check — concentrated in stress periods and on the overweight side.*

Everything the screening removed is preserved in the repository so the rejections can be reproduced: `./run.sh alldims` restores all three dimensions, `--regime` restores regime switching.

References

- Molchanov, A. & Stangl, J. (2024). The myth of business cycle sector rotation. *International Journal of Finance & Economics*.
- Moskowitz, T. & Grinblatt, M. (1999). Do industries explain momentum? *Journal of Finance*, 54, 1249–1290.
- Sarwar, G., Mateus, C. & Todorovic, N. (2020). Gauging the effectiveness of sector rotation strategies. *Journal of Asset Management*.
- FactSet. A practical approach to weighting signals.
- Newey, W. & West, K. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55, 703–708.

Research exercise. Not investment advice. Backtested results are hypothetical and do not represent actual trading.